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Project Title: Powering the Path: A Spatio-Temporal Analysis of Renewable Energy Infrastructure, Educational Investment, and Household Mobility in Victoria (2024–2026)

Introduction

Victoria's goal to reach 100% renewable energy by 2035 is already reshaping its economy and job locations. Investments in Renewable Energy Zones (REZs), transmission infrastructure, and regional Tech Schools are driving this shift [2], with solar power now contributing 16% of the target [1]. This raises an important question for my research: **how does the development of Victoria's REZs affect regional travel patterns and mobility, and does it promote sustainable travel or exacerbate transport inequality?**

Motivation

I am motivated to study how Victoria's transition from coal power to renewable energy through regional Renewable Energy Zones (REZs) reflects the broader global shift toward green energy [3]. While this transition creates new regional jobs, it may also change how people travel for work. Using 2024–2025 travel data, I will examine whether new renewable energy infrastructure supports sustainable travel or increases regional transport inequality.

Questions

1. In what ways does the spatial intensity of work and education-related travel align with the regional distribution of Victoria's new clean energy infrastructure and educational facilities?
2. How does the link between housing type, bicycle ownership, and proximity to major renewable energy facilities differ by socio-economic decile?
3. In what ways does the transition to regional energy hubs influence the accessibility of specialized vocational education?

Data source(s)

- **VISTA 2024-2025 (Relational):** A major survey of day-to-day travel diaries. It contains categorical (dweltype, origpurp1), ordinal (hhinc_group), and temporal (starthour) attributes.
 - **Trips** (~150k rows, 40 cols):
https://opendata.transport.vic.gov.au/dataset/174fbf40-bef0-434a-8af0-891c7f3d2323/resource/8d93c74-cf02-48d7-a368-fc207c0b3c4f/download/trips_vista_2024_2025.csv
 - **Persons** (~82k rows, 51 cols):
https://opendata.transport.vic.gov.au/dataset/174fbf40-bef0-434a-8af0-891c7f3d2323/resource/2a38732d-04dc-4af5-8a52-f281a8eceb7/download/person_vista_2024_2025.csv
 - **Households** (~32k rows, 30 cols):
https://opendata.transport.vic.gov.au/dataset/174fbf40-bef0-434a-8af0-891c7f3d2323/resource/1b73f806-1565-4243-96d7-78f9002c3e8f/download/household_vista_2024_2025.csv
- **State Budget Map Data 2023-24 (Spatial):** Contains spatial coordinates (lat, long) and quantitative (investment) attributes for ~1,000 infrastructure projects.
 - <https://discover.data.vic.gov.au/dataset/ed30724b-6ae7-4713-aff3-4f00b3666684/resource/71ca8b10-5a4e-483d-b3a3-a94421f4ef8b/download/state-budget-2023-24-budget-website-map-data.csv>
- **ABS SEIFA 2021 LGA Indexes (Tabular):** Provides ordinal Decile and Rank scores for 566 LGAs to evaluate socio-economic equity.
 - <https://www.abs.gov.au/statistics/people/people-and-communities/socio-economic-indexes-area-s-seifa-australia/2021/Local%20Government%20Area%2C%20Indexes%2C%20SEIFA%202021.xlsx>

References

- [1] Foley, E. (2026, February 12). *Solar energy chips in 16% of Victoria's renewable energy target*. PV Magazine Australia.
<https://www.pv-magazine-australia.com/2026/02/12/solar-energy-chips-in-16-of-victoria-renewable-energy-target/> [Accessed on: 15.03.2026 (checked 15.03.2026)].
- [2] VicGrid. (2025). *2025 Victorian Transmission Plan*. Department of Energy, Environment and Climate Action. <https://engage.vic.gov.au/victransmissionplan> [Accessed on: 15.03.2026 (checked 15.03.2026)].
- [3] Clean Energy Council. (2025). *Clean Energy Australia Report 2025*.
<https://cleanenergycouncil.org.au/news-resources/clean-energy-australia-report-2025> [Accessed on: 15.03.2026 (checked 15.03.2026)].